

CL-PFU database pipeline diagnostic report

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1 Introduction

This report provides diagnostics for the CL-PFU database pipeline. The purpose of this report is to compare key results from one version of the database to another.

2 Preliminaries

We first set the version of the database for which diagnostics are desired.

```
clpfu_version <- "v3.0a2"
```

Set data file paths.

```
world_data_file <- file.path("world_mats.rds")
```

We can download data using this chunk of code. To save time, this chunk is not normally evaluated when building the report (`eval=FALSE`). Run this chunk as needed.

```
# We first make a connection to the database.
conn <- PFUPipelineTools::get_scratchmdb_conn()
on.exit(DBI::dbDisconnect(conn))

# Next, download desired data and save locally for later use.
# We do this once to save time.
world_mats <- PFUPipelineTools::pl_filter_collect(
  version_string = clpfu_version,
  db_table_name = "PSUTReAllChopAllDsAllGrAll",
  Country == "World",
  conn = conn,
  collect = TRUE)

# Save to a file
world_mats |>
  saveRDS(file = world_data_file)

# Disconnect from the database management system.
DBI::dbDisconnect(conn)
```

Load data from the pre-saved file.

```
world_mats <- readRDS(file = world_data_file)
```

At this point, we are ready to make diagnostic graphs and tables.

3 Primary, final, and useful aggregations

3.1 Global primary and final exergy by sector

3.2 Global useful energy and exergy by energy carrier

The following figure is a recreation of Figure 7 of Brockway et al. (2024).

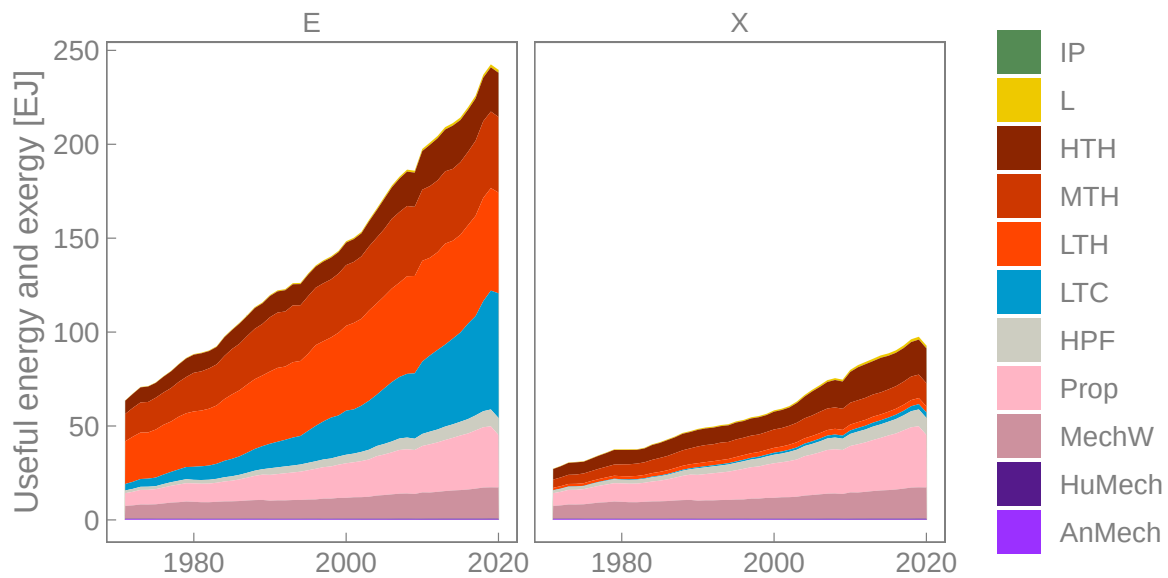


Figure 1: Recreation of Figure 7 of ?.

References

Brockway, Paul E, Matthew Kuperus Heun, Zeke Marshall, et al. 2024. "A Country-Level Primary-Final-Useful (CL-PFU) Energy and Exergy Database: Overview of Its Construction and 1971–2020 World-Level Efficiency Results." *Environmental Research: Energy* 1 (025005): 1–29. <https://doi.org/10.1088/2753-3751/ad4e39>.